BAYESIAN MODELING OF COMPLEXITY

Complexity, Modeling, & Bayes

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Tuesday 17th February, 2026

Hertie School of Public Governenance

Agenda

1 Why This Course?

2 Complex Systems

3 Modeling Paradigms

4 The Bayesian Connection

5 Course Roadmap

6 Wrap-Up

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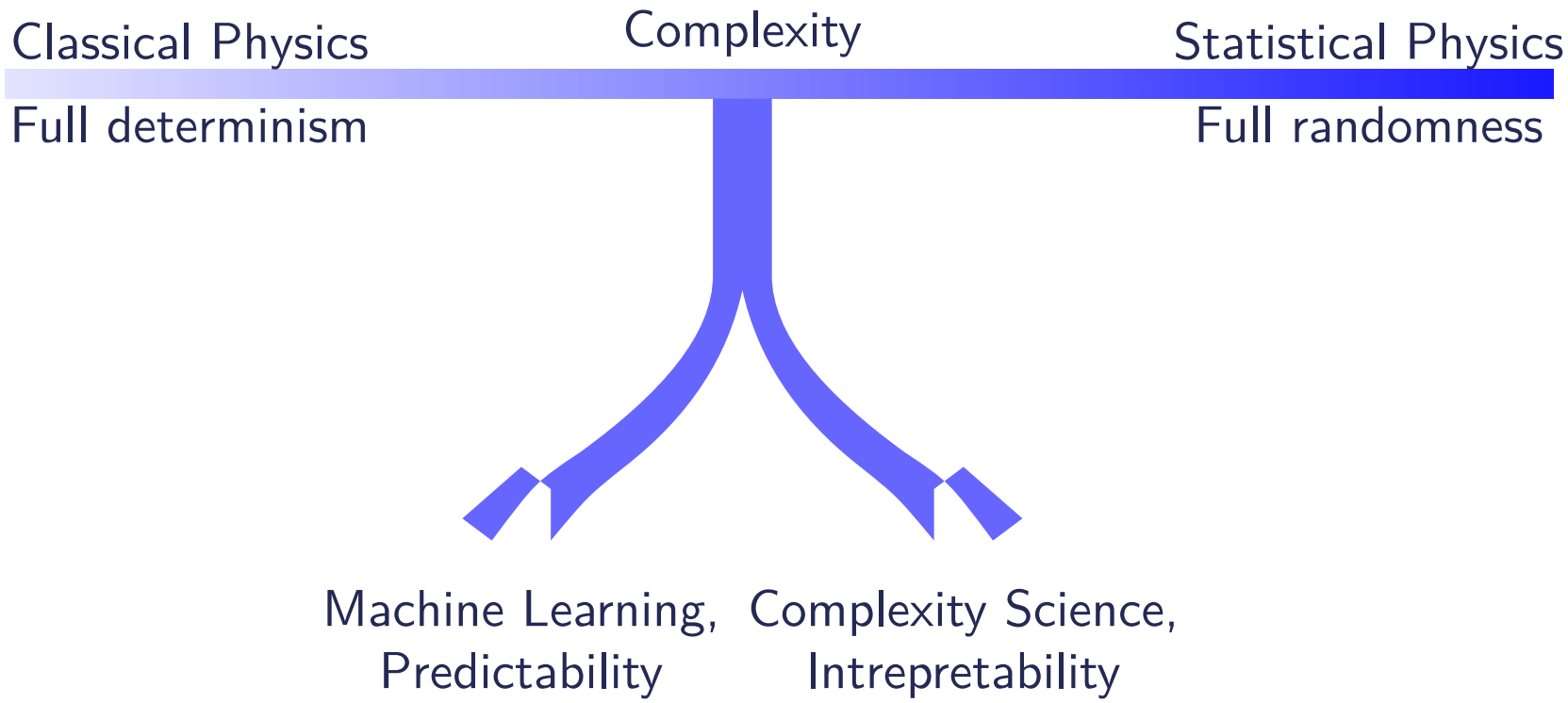
Where Does Complexity Science Sit?



Can we reconcile them?

Why Learning Complex Systems?

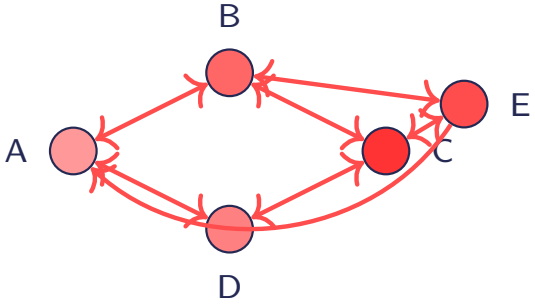
Traditional models assume:

- Linear relationships
- Homogeneous actors
- Equilibrium states
- Representative agents

Reality is often:

- Nonlinear and interdependent
- Heterogeneous and adaptive
- Far from equilibrium
- Context-dependent

Traditional
Linear, predictable



Complex
Heterogeneous,
feedback, adaptive

Key Insight

Many policy problems are fundamentally complex systems problems

Complex Systems & Policy?

Policy domains as complex systems:

- **Public health:** Disease spread, vaccination campaigns
- **Climate policy:** Emissions, adaptation, behavior change
- **Urban planning:** Housing, transportation, segregation
- **Economic policy:** Markets, inequality, innovation
- **Education:** School choice, peer effects, outcomes

Why this matters:

- Interventions have unexpected consequences
- One-size-fits-all policies often fail
- Need to understand micro-macro links

Interventions create cascading effects

Course Goal

Build tools to understand, model, and learn from complex systems

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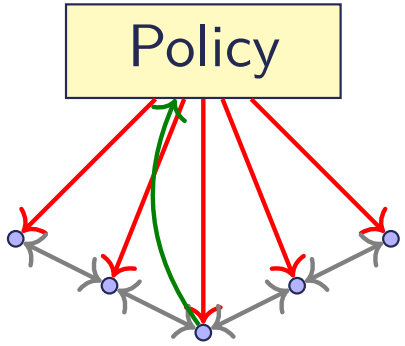

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Motivating Problem: Residential Segregation

Observation:

- Cities show strong segregation by race/ethnicity
- Persists despite anti-discrimination laws
- Traditional models struggle to explain

Segregated city

Questions:

- What mechanisms drive this?
- How strong are individual preferences?
- What interventions might help?

Traditional approach: Differential equations with the assumption of homogeneity

Our approach: ABM to observe micro mechanisms.

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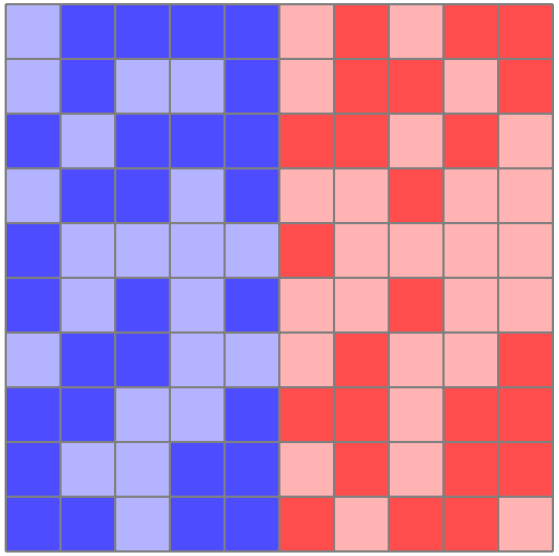

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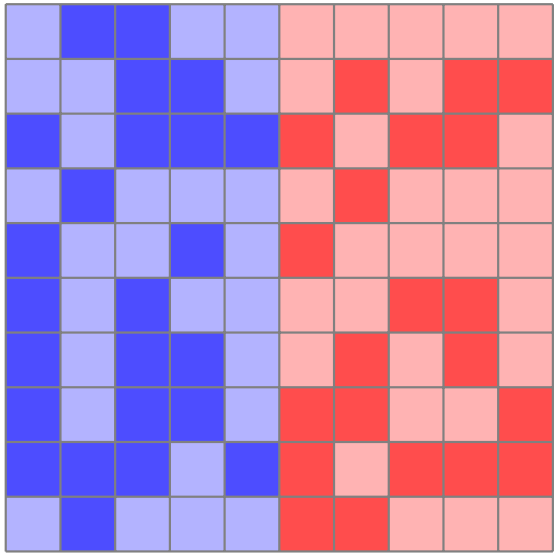

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What Makes a System "Complex"?

Complex System: A system in which large network of simple components with no central control and simple rules of operation give rise to complex collective behavior, sophisticated information processing, and adaptation via learning or evolution [1].

Examples across domains:

- **Biology:** Immune system, brain
- **Social:** Markets, cities, social networks
- **Physical:** Climate, traffic, power grids
- **Technological:** Internet, supply chains

Common features:

- Many components
- Local interactions
- Nonlinear dynamics
- Adaptation/learning
- No central control

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Understanding individual components $\neq$ understanding system behavior

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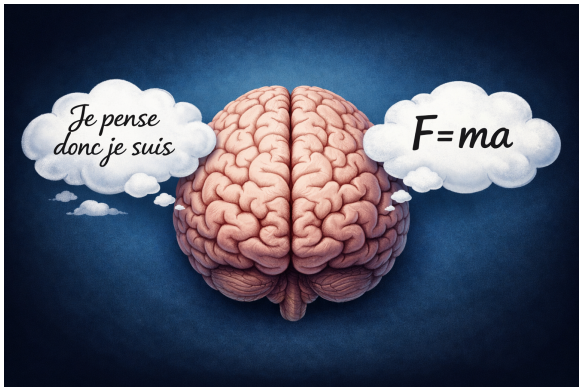
Property 1: Emergent Behavior

System-level patterns that cannot be predicted from individual rules.

Examples:

- **Bird flocking:** Coordinated movement from simple rules (align, avoid, cohere)
- **Traffic jams:** No accidents needed, density alone creates waves
- **Markets:** Prices emerge from individual trades
- **Consciousness:** From neuron firing patterns



Emergent Phenomena.

The Whole is More Than the Sum of Its Parts

Emergent properties exist only at the system level, i.e., you cannot study a single bird to understand flocking

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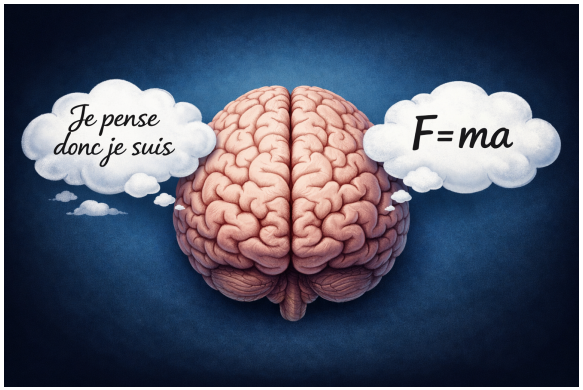

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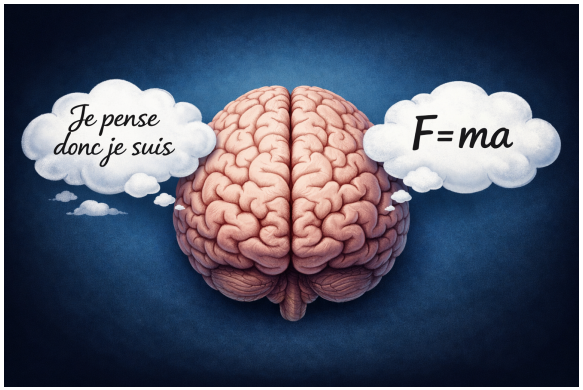

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Property 2: Self-Organization

Self-Organization: The capacity of a system to make its own structure more complex [2].

Properties:

- **The freedom to experiment** is necessary for being self-organized → new way of doing things, require experimentation.
- **Simple rules** can lead to self-organization property → evolution rules are quite simple: diversification, selection, amplification.

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Ant colonies

The ant colony restructures itself to respond to a challenge in the environment.

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Property 3: Adaptation

Components learn, evolve or adjust behavior based on experience

Examples:

- **Immune system:** Learns to recognize pathogens
- **Evolution:** Species adapt to environment
- **Markets:** Traders adjust strategies

Mechanisms:

- Variation (trying different things)
- Selection (success/failure)
- Retention (memory)

Dynamic Fitness Landscape

Individual components change behavior as the system learns (unlike static physical systems).

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Nonlinearity: Small causes can have large effects (and vice versa)

Feedback loops: Outputs influence inputs

- **Positive feedback:** Amplification (Reinforcing)
- **Negative feedback:** Stabilization (Balancing)

Examples:

- **Markets:** Panic selling
  Price drop → fear → more selling → bigger drop
- **Epidemics:** Exponential growth
  More infected → more contacts → more infected

Butterfly Effect

Sensitive dependence on initial conditions, i.e., tiny changes can cascade into vastly different outcomes

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Example: Logistic Map (Rabbit Population)

Model: $x_{n+1} = r \cdot x_n(1 - x_n)$

Simple rule:

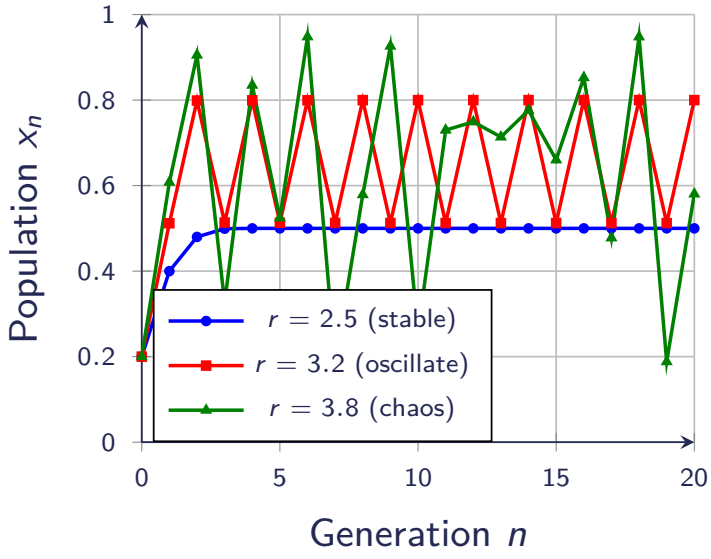
- x_n: population fraction (0 to 1)
- r: growth rate parameter
- Nonlinear due to $(1 - x_n)$ term

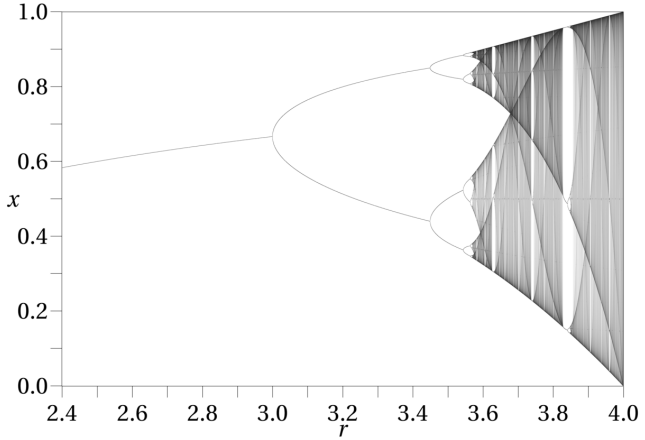
Behavior depends on r:

- $r < 3$: Stable equilibrium
- $3 < r < 3.57$: Oscillations
- $r > 3.57$: Chaos!

Simple equation → complex dynamics

Live Interaction





Key Lesson

Perfect deterministic system can lead to chaotic begaviour.

Example: Mandelbrot Set

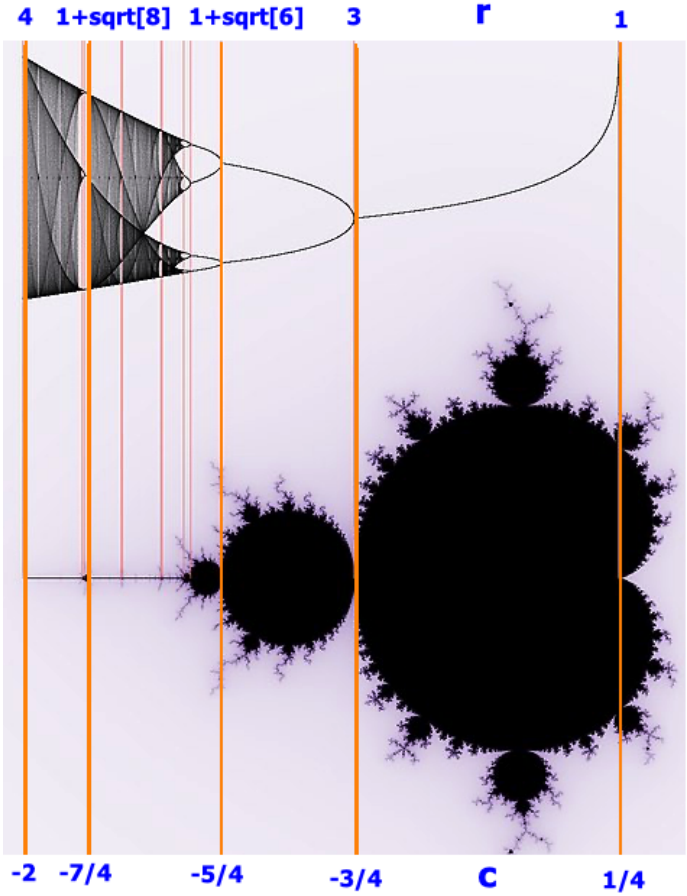
Iteration rule: $z_{n+1} = z_n^2 + c$ (z is a complex number)

How it works:

- Start with $z_0 = 0$
- Pick a complex number c
- Iterate: does $|z_n|$ stay bounded?
- If yes: c is in the set (black)
- If no: c escapes (colored)

Remarkable properties:

- Infinite detail at all scales
- Self-similar patterns
- Boundary has fractal dimension



Mandelbrot set & logistic map 3D View

Simple local rules → emergent global structure (no blueprint!)

Identify Feedback Loops

Scenario: Urban gentrification process

1. **Individual work:** On your own, identify at least 2 feedback loops (positive or negative) in this system:
   - Artists move to cheap neighborhood
   - Neighborhood becomes "trendy"
   - Coffee shops and galleries open
   - Property values increase
   - Original residents priced out
   - Artists leave (now too expensive)
2. What loops do you find (negative or positive)?

Hint: Look for circular causation: A influences B influences C influences A

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Positive feedback loops (amplification):

1. Trendiness → more amenities → more demand → higher prices → more investment → more trendiness
2. Price increase → displacement → demographic shift → more luxury demand → more price increase

Negative feedback loops (limits):

1. High prices → artists leave → less cultural activity → less trendiness → demand softens
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Complex systems have **multiple interacting feedbacks**. Understanding the system requires tracking these circular causations, not just linear cause-effect

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Challenges with complex systems:

- Real-world experiments often impossible (ethics, scale, cost)
- Controlled manipulation difficult
- Prediction requires understanding mechanisms

Models allow us to:

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2. **Explore scenarios:** How does system behave under different conditions?
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Goal: Not a perfect representation, but a *useful* simplification that captures key mechanisms

Simple models	Complex models
Learning the mechanisms	Predictive power

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Equation-Based Models (Top-Down)

Idea: Describe system with differential equations at aggregate level

Example: SIR Epidemic Model

$$\frac{dS}{dt} = -\beta SI, \qquad \frac{dI}{dt} = \beta SI - \gamma I, \qquad \frac{dR}{dt} = \gamma I$$

- S: Susceptible population
- I: Infected population
- R: Recovered population
- β: Transmission rate
- γ: Recovery rate

Strengths:

- Analytically tractable
- Fast to simulate
- Well-understood mathematics

Limitations:

- Assumes homogeneous mixing
- No individual variation
- Difficult for spatial/network structure

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Approach: Simulate individual agents and their interactions → system behavior emerges **Core components:**

1. **Agents:** Autonomous entities with:
   - State (properties, memory)
   - Behavior rules (decision-making)
   - Local perception
2. **Environment:** Space where agents interact
   - Can be spatial (grid, network) or abstract
3. **Interactions:** How agents affect each other
   - Often local (neighbors, network connections)

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Define rules at individual level → observe emergent patterns at system level

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ABM Example: Schelling Segregation Model

Question: Can strong segregation emerge from mild individual preferences?

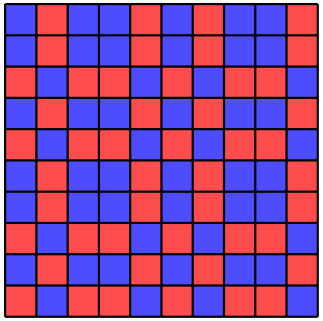
Setup:

- Agents on a grid (2 types: blue, red)
- Each has tolerance threshold (e.g., 30%)
- "Happy" if $\geq$ 30% neighbors are same type
- "Unhappy" agents move to random empty cell

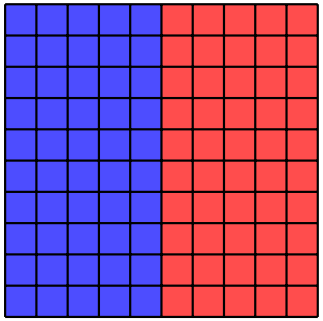
Simple rule:

- If unhappy $\rightarrow$ move
- If happy $\rightarrow$ stay

Initial (random)



After 20 steps



Emergent Result

Mild preference (30%) $\rightarrow$ Strong segregation (80%+)! **We'll build this next session.**

ABM Example: Schelling Segregation Model

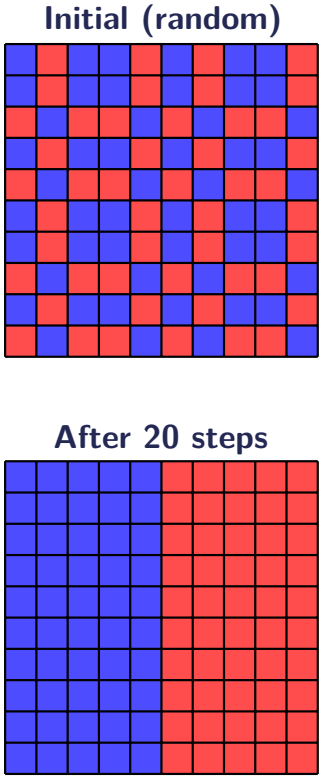
Question: Can strong segregation emerge from mild individual preferences?

Setup:

- Agents on a grid (2 types: blue, red)
- Each has tolerance threshold (e.g., 30%)
- "Happy" if $\geq$ 30% neighbors are same type
- "Unhappy" agents move to random empty cell

Simple rule:

- If unhappy $\rightarrow$ move
- If happy $\rightarrow$ stay



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When to Use Agent-Based Models?

ABMs are particularly useful when:

1. **Heterogeneity matters**
   - Individual differences drive system behavior
   - Example: Varied individuals' perceptions drive the prices
2. **Local interactions are key**
   - Agents interact with neighbors, not the entire population.
   - Example: Disease spread in contact networks
3. **Spatial/network structure matters**
   - Location or network position affects outcomes
   - Example: Urban sprawl, information diffusion
4. **Adaptive behavior present**
   - Agents learn or change strategies
   - Example: Bayesian estimation of a certain parameter
5. **Emergent phenomena of interest**
   - System-level patterns from micro-level rules
   - Example: Traffic jams, market crashes

Equation-Based vs. Agent-Based Models

Feature	Equation-Based	Agent-Based
Level	Aggregate (top-down)	Individual (bottom-up)
Heterogeneity	Difficult to model	Natural to include
Space/Network	Requires special treatment	Built-in
Analysis	Analytical tools available	Primarily computational
Computational	Fast	Can be slow
Emergence	Harder to capture	Designed for it
Calibration	Standard methods	Challenging
Validation	Established protocols	Active research area

Complementary Approaches

Not "either/or", each useful for different questions. Sometimes hybrid models combine both.

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The Challenge: Learning from Data

We can build flexible ABMs, but:

- ABMs have parameters (e.g., tolerance threshold in Schelling)
- How do we set these parameters?
- How do we learn from observed data?

Example questions:

- What tolerance level explains observed segregation in Chicago?
- How infectious is this new pathogen (based on early outbreak data)?
- What are citizens' climate policy preferences (from survey + behavior)?

The Problem

ABMs often have no closed-form likelihood $P(\text{data}|\text{parameters})$. Traditional statistical methods (maximum likelihood, least squares) struggle with ABMs.

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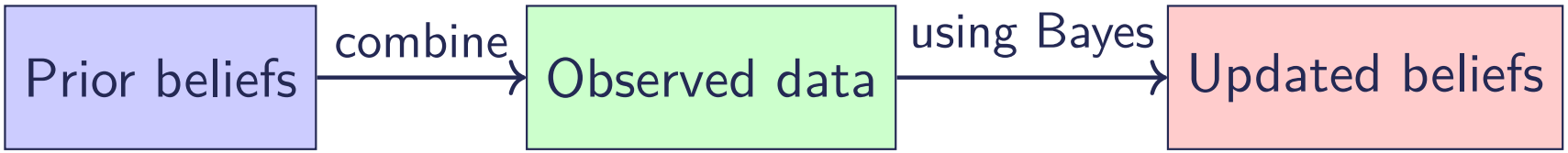
Bayesian Thinking: High-Level Preview

Core idea: Update beliefs about parameters based on evidence



Mathematical form:

$$\underbrace{P(\theta|\text{data})}_{\text{Posterior}} \propto \underbrace{P(\text{data}|\theta)}_{\text{Likelihood}} \times \underbrace{P(\theta)}_{\text{Prior}}$$

- **Prior:** What we believe before seeing data
- **Likelihood:** How probable is the data for given parameters?
- **Posterior:** Updated beliefs after seeing data

Details in sessions 4-6, for now, just the intuition

Why Bayes + ABM?

Bayesian methods are a good fit for ABMs because:

1. **Quantify uncertainty**: Not just point estimates, but also full distributions over parameters. **Example**: Tolerance is likely between 25-35% (95% credible interval).
2. **Incorporate prior knowledge**: Domain expertise, previous studies. Especially valuable when data is limited.
3. **Flexible calibration methods**: Approximate Bayesian Computation (ABC) for models without tractable likelihood → Match patterns instead of exact data
4. **Natural for sequential learning**: Today's posterior = tomorrow's prior. We update as new data arrives.

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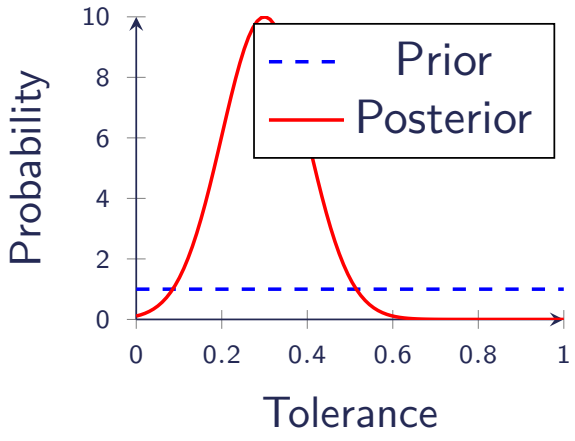
Example: Calibrating Schelling with Bayes

Scenario: We observe real segregation patterns in a city

Question: What tolerance threshold best explains this pattern?

Approach:

1. **Prior:** Tolerance $\sim$ Uniform(0, 1)
   We're uncertain,any value plausible
2. **Simulate:** Run Schelling for many tolerance values
3. **Compare:** Which runs match observed segregation?
4. **Posterior:** Distribution over tolerance that fits data



Prior (flat) $\rightarrow$ Posterior (peaked)

Result: "Data suggests tolerance $\approx$ 30% (with uncertainty quantified)"

We will talk about it more in session 8!

Course Roadmap

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12-session Journey

Sessions	Topic	Key Deliverable
1-3	Complexity & ABM	None
4-6	Bayesian Reasoning	Midterm Break Repo + ODD Protocol
7	Bayesian Linear Models	None
8	Guest: TBD	None
9	Sensitivity Analysis	None
10-11	Hierarchical Bayesian Modeling	None
12	Final Presentations	Full Project Repo and Presentation
14		Analysis Paper

The General Flow

Build complex system models → **Learn** Bayesian inference → **Combine** to learn from data and ABMs

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Next Session: ODD Protocol & Schelling Implementation

Session 2:

- ODD framework: How to design and document ABMs
- Start building Schelling model (live coding)
- Grid setup, agent initialization

Session 3:

- Complete Schelling implementation
- Pattern-oriented modeling
- Parameter sweeps and analysis
- **ABM project chosen**

To prepare:

- Review basics of OOP in Python: classes, methods, objects etc.

Questions?

See you next session to start building!

References

[1] M. Mitchell, *Complexity: A guided tour*. Oxford university press, 2009.

[2] D. Meadows, *Thinking in systems: International bestseller*. chelsea green publishing, 2008.